UI/UX DESIGN BRIEF & USER EXPERIENCE SPECIFICATION

Human-Centered Design, Interface Architecture, and Data Visualization for Heavy Oil Digital Twin

Heavy Oil Wells – Baghewala Field, Rajasthan

Smart India Hackathon (SIH) 2026

UI/UX Design System Specification

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1 Design Vision & UX Philosophy

1.1 Industrial Control Room Metaphor (ISA-101 Compliance)

The UI/UX design for the Baghewala Well-to-Surface Digital Twin bridges complex subsurface thermal physics, downhole elastic wave mechanics, and surface VFD control into an intuitive, high-density, mission-critical operational cockpit. Grounded in the **ANSI/ISA-101.01 Human Machine Interfaces (HMI)** standards, the UI prioritizes situational awareness, cognitive load reduction, and instant anomaly recognition.

1.2 Core Design Principles

1. **Dark-Mode First Architecture**: Optimized for 24/7 SCADA control rooms and outdoor ruggedized field tablets. Dark backgrounds (#121820) reduce glare, conserve display power, and make color-coded operational hazards pop instantly.
2. **Actionable Physics Visualizations**: Complex numerical equations (such as 1D Damped Wave dynamics and Andrade thermal viscosity curves) are rendered as live, interactive graphics rather than raw tabular data.
3. **Contextual Color Semantics**: Color is reserved strictly for operational status and hazard communication. Muted slate tones represent normal baseline states, while bright signal colors highlight critical physics anomalies (e.g., Rod Float, Impact Loading, Fluid Pound).
4. **Multi-Level Progressive Disclosure**: Information flows logically from field-wide macro-metrics down to individual downhole plunger force-displacement dynamics.

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2 Target User Personas & Use Cases

User Persona	Primary Goal	Key UX Pain Point	Tailored UI Solution
Field Operator *(At Wellhead)*	Daily inspection, manual setpoint checks, quick fault resets	Clunky legacy SCADA dialogs, poor daylight visibility	High-contrast "Field Mode", 48px+ touch targets, instant VFD auto/manual override switches.
Production Engineer *(Control Room)*	Optimize daily production, avoid rod breakage, monitor SPM	Disconnected dyno cards and thermal cooling schedules	Live Surface vs. Downhole Dyno Card overlay, real-time SPM_{safe} limits, automated rod float alerts.
EOR Specialist *(Reservoir Team)*	Track steam efficiency, evaluate thermal decay $T(t)$, plan CSS	Lack of insight into when a well has cooled too far	Interactive exponential cooling curve $T(t)$, automatic $dJ/dt \leq 0$ economic cutoff notification.
Asset Operations Manager	Monitor field revenue, minimize overall SOR, track OPEX	Tabular data overload, missing macro trends	Unified GIS asset map, fleet-wide SOR heatmaps, energy intensity gauges.

Table 1: User Persona Analysis and UI Customization Matrix

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3 Information Architecture (IA) & Navigation Map

3.1 Hierarchical Navigation Topology

The user interface is structured into three primary depth levels:

- **Level 1: Asset Fleet Command Center (GIS Overview)**
  - Spatial distribution of Baghewala wells over Jodhpur Sandstone formation.

    - Global KPI Header: Total Field Oil Rate (bpd), Average SOR, Total Energy Consumption (kWh/bbl), Active Rod Float Warnings.

- **Level 2: Well-to-Surface Cockpit (Single Well Deep-Dive)**
    - *Module A: Subsurface Thermal & Fluid Dynamics* ($T(t)$, $\mu(t)$, Steam Enthalpy).
    - *Module B: Downhole Wave Mechanics & Dyno Card Studio* (1D Wave, Surface vs. Plunger Card, $PPRL$).
    - *Module C: Closed-Loop VFD Control Center* (SPM_{safe}, Auto-Throttle, Override).
    - *Module D: Predictive Diagnostics & Stress Profile* (ResNet classification, Rod Compression).
    - *Module E: EOR Lifecycle Economics* (Cumulative Profit $J(t)$, dJ/dt cutoff trigger).

- **Level 3: Operational Logs, Calibration & Alarm Management**

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4 Detailed Screen Specifications

4.1 Screen 1: Well-to-Surface Cockpit (Core Interactive Interface)

The primary operational workspace combines physical schematics with live telemetry graphs in a split-screen dashboard:

Layout Quadrant	Visual Components & Data Displays	User Interactions & Controls
Left Panel: Subsurface & Surface Schematic	Animated 2D/3D cutaway of Sucker Rod Pump, polished rod, tubing string, downhole plunger, and reservoir thermal zone. Dynamic color fill indicates reservoir temperature gradients.	Hover over components to reveal live stress (σ), depth (x), and fluid temperature (T).
Top Right: Dynamometer Studio	WebGL Canvas overlaying **Surface Dyno Card** (Cyan line) and **Reconstructed Downhole Plunger Card** (Green/Red line). Live dot travels along the card loop in real-time (60 FPS).	Toggle historical reference cards; overlay theoretical pump fillage envelopes; zoom in on valve opening points.
Middle Right: Thermal Decay & Viscosity	Dual-axis line plot showing predicted exponential temperature decay $T(t)$ against crude viscosity $\mu(t)$ surge. Includes SPM_{safe} threshold line.	Drag production time slider (t days) into the future to simulate cooling and predicted viscosity surge.
Bottom Right: VFD Control Console	Large SPM_{safe} gauge with color-coded safety zones. Real-time SPM slider, VFD target Hz display, and "Auto-Throttle" toggle button.	Slide to manually adjust SPM; toggle AI closed-loop control; trigger emergency safe idle (2.0 SPM).

Table 2: Well Cockpit UI Component Layout Specification

4.2 Screen 2: EOR Cycle Economics & Cutoff Predictor

Designed specifically for Reservoir Engineers to evaluate when to stop pumping (Puff phase) and initiate the next steam injection (Huff phase):

- **Economic Objective Gauge**: Real-time gauge displaying rate of profit change dJ/dt.
- **State Indicators**: Green state when $dJ/dt > 0$; blinking Amber when $dJ/dt \to 0$; Red alert when $dJ/dt \leq 0$ (Trigger Next CSS Cycle).
- **SOR Lifecycle Chart**: Cumulative Steam-Oil Ratio curve ($m^3_{\text{steam}}/m^3_{\text{oil}}$) tracking economic feasibility limits.

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5 Design System & Visual Style Guide

5.1 Color Palette (ISA-101 Compliant Dark Theme)

Color Name	Hex Code	RGB	UI Purpose & Semantic Application
Obsidian Background	`#121820`	(18, 24, 32)	Global app background (reduces eye fatigue).
Slate Panel Surface	`#222E40`	(34, 46, 64)	Card and container backgrounds, panel dividers.
Electric Cyan	`#00E5FF`	(0, 229, 255)	Surface dyno card, active telemetry lines, primary focal points.
Emerald Green	`#00E676`	(0, 230, 118)	Normal downhole dyno card, healthy equipment status.
Amber Hazard	`#FFAB00`	(255, 171, 0)	Viscosity surge warning, high motor load, approaching SPM_{safe}.
Crimson Alert	`#FF1744`	(255, 23, 68)	Rod float detected, negative axial load, pump unsetting, $dJ/dt \leq 0$.
Steam Enthalpy Violet	`#D500F9`	(213, 0, 249)	Steam injection phase indicators, thermal heat maps.

Table 3: System Color Palette and Operational Semantics

5.2 Typography & Numerical Telemetry Guidelines

- **Primary Sans-Serif (UI & Labels)**: `Inter` or `Roboto` (Clean, high legibility at small point sizes).
- **Monospace (SCADA Data & Coordinates)**: `JetBrains Mono` or `Fira Code` (Enforces fixed-width tabular numbers to prevent UI jump during high-frequency live SCADA updates).

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6 Accessibility, Responsiveness & Touch Adaptability

6.1 Field Mobile / Tablet Mode (Outdoor Adaptability)

Field technicians operating near noisy, high-vibration pumpjacks require specialized UI accommodations:

- **High-Contrast Daylight Mode**: One-tap toggle that switches dark slate surfaces to high-contrast white/light-grey for outdoor sunlight readability.
- **Touch Target Enforcement**: Minimum interactive button dimensions set to 48×48 px with 12 px spacing to support usage with heavy industrial safety gloves.
- **Multi-Sensory Haptic & Audio Feedback**: Critical events (e.g., severe rod compression during downstroke) trigger distinct audio alarms and tablet vibration pulses.